

README

HelixCode - Comprehensive User Manual

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1. Introduction

1.1 What is HelixCode?

HelixCode is an enterprise-grade distributed AI development platform that revolutionizes software development through intelligent automation. It combines the power of 14+ leading AI providers with advanced tooling, distributed computing, and sophisticated workflow automation.

Core Philosophy: Empower developers with AI assistance while maintaining full control and transparency.

1.2 Key Features

AI Integration (14+ Providers)

- **Premium Providers:** Claude, Gemini, OpenAI, AWS Bedrock, Azure OpenAI, VertexAI, Groq, Mistral
- **Free Providers:** XAI Grok, OpenRouter, GitHub Copilot, Qwen
- **Local Models:** Ollama, Llama.cpp (100% offline)
- **Advanced Features:** Extended thinking, prompt caching, 2M token contexts

Smart Tools

- **File System:** Intelligent caching, atomic operations, pattern matching
- **Shell:** Sandboxed execution with security controls
- **Browser:** Chrome automation for testing and scraping
- **Web:** Multi-provider search and HTML parsing
- **Voice:** Hands-free coding with Whisper transcription
- **Mapping:** Tree-sitter AST parsing for 30+ languages

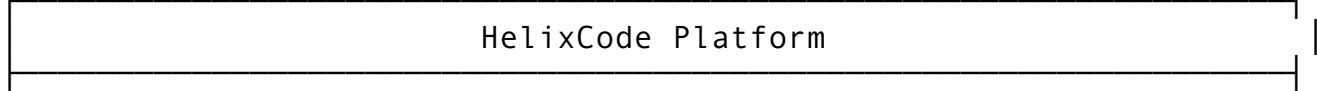
Intelligent Workflows

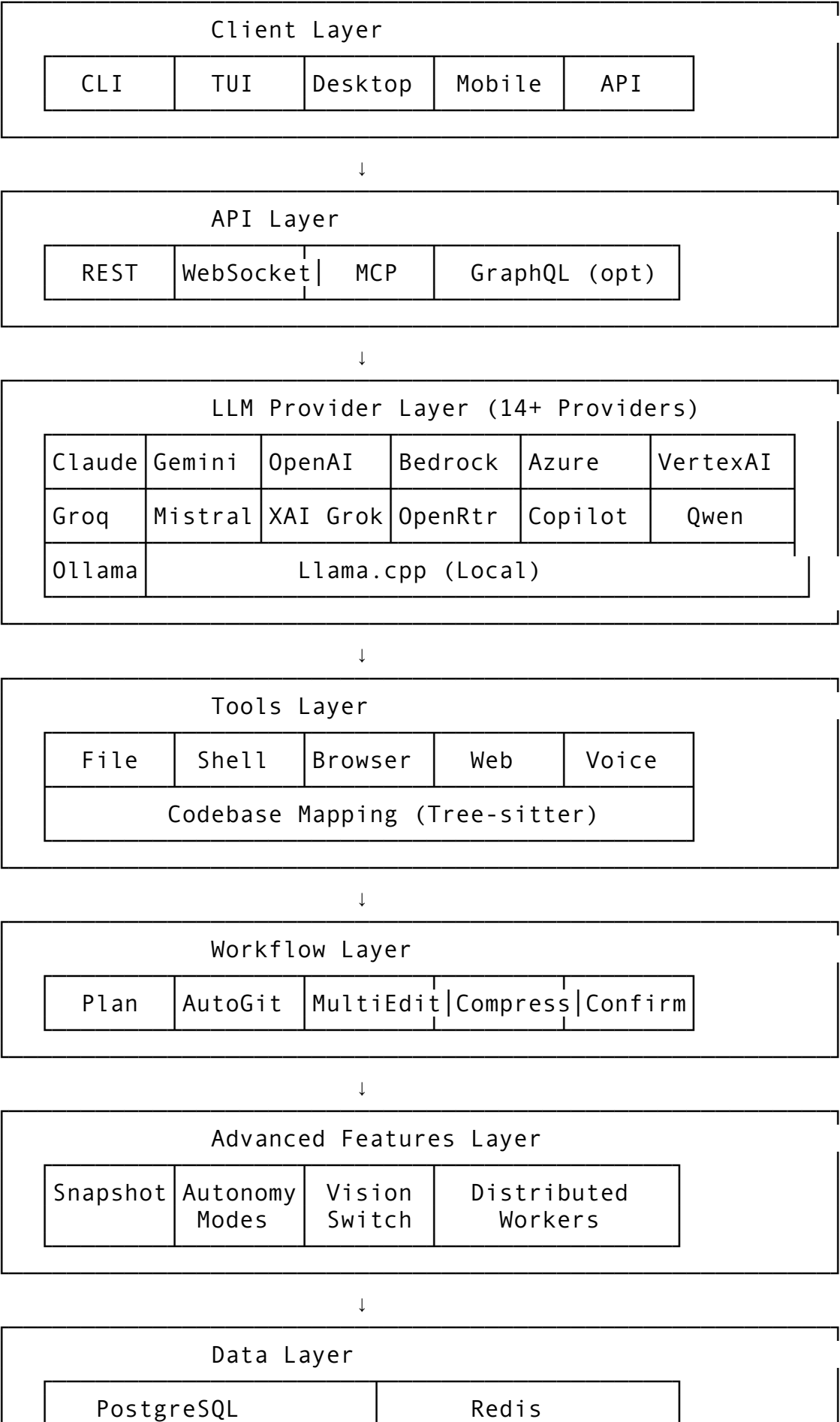
- **Plan Mode:** Two-phase plan-then-execute workflow
- **Auto-Commit:** LLM-generated semantic commit messages
- **Multi-File Edit:** Atomic transactions across multiple files
- **Context Compression:** Automatic conversation summarization
- **Tool Confirmation:** Interactive approval for dangerous operations

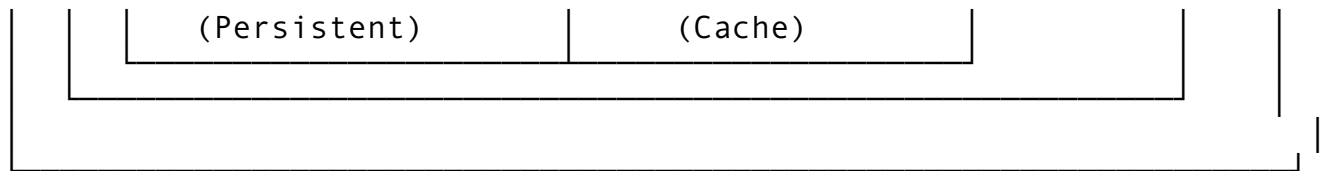
Enterprise Features

- **Snapshots:** Git-based checkpoint system with instant rollback
- **Autonomy Modes:** 5 levels of AI automation (None → Full Auto)
- **Vision Auto-Switch:** Automatic model switching for images
- **Distributed Workers:** SSH-based worker pools with auto-installation
- **MCP Protocol:** Full Model Context Protocol implementation

1.3 Architecture Overview







1.4 Use Cases

Software Development

- Full-stack application development
- Code refactoring and optimization
- Automated testing and bug fixing
- Documentation generation
- Code review automation

DevOps & Infrastructure

- Infrastructure as Code (IaC) generation
- CI/CD pipeline automation
- Configuration management
- Monitoring and alerting setup

Data Engineering

- ETL pipeline development
- Data analysis automation
- SQL query optimization
- Schema design and migration

Research & Prototyping

- Rapid prototyping
- Algorithm implementation
- Performance benchmarking
- Literature review automation

Enterprise Development

- Microservices architecture
 - Legacy code modernization
 - Compliance and security auditing
 - Multi-team collaboration
-

2. Installation

2.1 System Requirements

Minimum Requirements

- **OS:** Linux (Ubuntu 20.04+), macOS (11+), Windows (10+/WSL2)
- **CPU:** 2 cores
- **RAM:** 4 GB
- **Storage:** 10 GB
- **Go:** 1.24.0 or later
- **Git:** 2.30+

Recommended Requirements

- **CPU:** 4+ cores
- **RAM:** 8+ GB
- **Storage:** 20+ GB SSD
- **PostgreSQL:** 14+
- **Redis:** 7+ (optional, recommended for production)

For Production

- **CPU:** 8+ cores
- **RAM:** 16+ GB
- **Storage:** 50+ GB SSD
- **PostgreSQL:** 15+ with replication
- **Redis:** 7+ with persistence
- **Load Balancer:** Nginx, HAProxy, or cloud LB

2.2 Quick Start Installation

```
# 1. Clone repository
git clone https://github.com/your-org/helixcode.git
cd helixcode/HelixCode
```

```
# 2. Install dependencies
go mod download
```

```
# 3. Generate assets (logo processing)
make logo-assets
```

```
# 4. Build all components
make build
```

```
# 5. Run tests to verify installation
make test
```

```
# 6. Start development server
make dev
```


Verify Installation:

```
# Check server health
curl http://localhost:8080/health

# Expected response:
# {"status":"healthy","version":"2.0","uptime":"10s"}
```

2.3 Production Deployment

Step 1: Database Setup

```
# Install PostgreSQL
# Ubuntu/Debian
sudo apt-get install postgresql-15

# macOS
brew install postgresql@14

# Create database and user
sudo -u postgres psql

CREATE DATABASE helixcode;
CREATE USER helixcode WITH PASSWORD 'your_secure_password';
GRANT ALL PRIVILEGES ON DATABASE helixcode TO helixcode;
\q
```

Step 2: Redis Setup (Optional but Recommended)

```
# Install Redis
# Ubuntu/Debian
sudo apt-get install redis-server

# macOS
brew install redis

# Configure Redis
sudo nano /etc/redis/redis.conf

# Set password (uncomment and modify):
# requirepass your_secure_redis_password

# Start Redis
sudo systemctl start redis
sudo systemctl enable redis
```

Step 3: Environment Configuration

```
# Copy example environment file
cp .env.example .env
```

```
# Edit environment variables
nano .env
```

Required Variables (.env):

```
# Authentication
HELIX_AUTH_JWT_SECRET=your-super-secure-jwt-secret-minimum-32-
    characters
```

```
# Database
HELIX_DATABASE_HOST=localhost
HELIX_DATABASE_PORT=5432
HELIX_DATABASE_NAME=helixcode
HELIX_DATABASE_USER=helixcode
HELIX_DATABASE_PASSWORD=your_secure_password
```

```
# Redis
HELIX_REDIS_HOST=localhost
HELIX_REDIS_PORT=6379
HELIX_REDIS_PASSWORD=your_secure_redis_password
```

```
# Server
HELIX_ENV=production
HELIX_SERVER_PORT=8080
```

```
# AI Providers (at least one required)
ANTHROPIC_API_KEY=sk-ant-your-key-here
GEMINI_API_KEY=your-gemini-key
OPENAI_API_KEY=sk-your-openai-key
```

Step 4: Build Production Binaries

```
# Build optimized production binaries
make prod
```

```
# This creates binaries in bin/:
# - bin/helixcode-server (main server)
# - bin/helixcode-cli (CLI client)
# - bin/helixcode-tui (terminal UI)
```

Step 5: Run Production Server

```
# Run server with production config
./bin/helixcode-server --config config/config.yaml
```

```
# Or use systemd service
sudo cp scripts/systemd/helixcode.service /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl start helixcode
sudo systemctl enable helixcode

# Check status
sudo systemctl status helixcode
```

2.4 Docker Deployment

Recommended for production: Docker Compose provides complete stack.

Step 1: Prepare Environment

```
# Create production .env file
cp .env.example .env
nano .env

# Set all required variables (see section 2.3)
```

Step 2: Deploy with Docker Compose

```
# Start all services
docker-compose up -d

# Services included:
# - helixcode-server (main application)
# - postgres (database)
# - redis (cache)
# - nginx (reverse proxy)
# - prometheus (monitoring)
# - grafana (dashboards)
```

Verify Deployment:

```
# Check all services are running
docker-compose ps
```

```
# Expected output:
# NAME                                STATUS
# helixcode-server                    Up
# helixcode-postgres                  Up
# helixcode-redis                     Up
# helixcode-nginx                     Up
# helixcode-prometheus                Up
# helixcode-grafana                   Up
```

```
# Test application
curl http://localhost/health

# View logs
docker-compose logs -f helixcode-server
```

Step 3: Access Services

- **Application:** http://localhost
- **API:** http://localhost/api
- **Grafana:** http://localhost:3000 (admin/admin)
- **Prometheus:** http://localhost:9090

2.5 Kubernetes Deployment

For large-scale production deployments with high availability.

Prerequisites

- Kubernetes cluster (1.24+)
- kubectl configured
- Helm 3+ (optional)

Step 1: Create Namespace

```
kubectl create namespace helixcode
```

Step 2: Create Secrets

```
# Database credentials
kubectl create secret generic helixcode-db \
  --from-literal=password=your_db_password \
  -n helixcode

# JWT secret
kubectl create secret generic helixcode-auth \
  --from-literal=jwt-secret=your_jwt_secret \
  -n helixcode

# AI provider API keys
kubectl create secret generic helixcode-llm \
  --from-literal=anthropic-key=sk-ant-... \
  --from-literal=gemini-key=... \
  --from-literal=openai-key=sk-... \
  -n helixcode
```

Step 3: Deploy Database

```
# Apply PostgreSQL StatefulSet
kubectl apply -f k8s/postgres.yaml -n helixcode

# Wait for database to be ready
kubectl wait --for=condition=ready pod -l app=postgres -n
    helixcode
```

Step 4: Deploy Application

```
# Apply all manifests
kubectl apply -f k8s/helixcode.yaml -n helixcode

# Components deployed:
# - Deployment (3 replicas)
# - Service (LoadBalancer)
# - ConfigMap (config.yaml)
# - HorizontalPodAutoscaler (2-10 pods)
```

Step 5: Verify Deployment

```
# Check pods
kubectl get pods -n helixcode

# Check services
kubectl get svc -n helixcode

# Get external IP
kubectl get svc helixcode -n helixcode -o
    jsonpath='{.status.loadBalancer.ingress[0].ip}'

# Test health endpoint
curl http://<EXTERNAL-IP>/health
```

2.6 Platform-Specific Setup

Linux

```
# Ubuntu/Debian
sudo apt-get update
sudo apt-get install -y build-essential git postgresql-15 redis-
    server

# Fedora/RHEL
sudo dnf install -y gcc git postgresql-server redis

# Install Go
wget https://go.dev/dl/go1.26.3.linux-amd64.tar.gz
```

```
sudo tar -C /usr/local -xzf go1.26.3.linux-amd64.tar.gz
echo 'export PATH=$PATH:/usr/local/go/bin' >> ~/.bashrc
source ~/.bashrc
```

macOS

```
# Install Homebrew (if not installed)
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/
    Homebrew/install/HEAD/install.sh)"
```

```
# Install dependencies
brew install go postgresql@14 redis git
```

```
# Start services
brew services start postgresql@14
brew services start redis
```

Windows (WSL2)

```
# Install WSL2 with Ubuntu
wsl --install -d Ubuntu
```

```
# Inside WSL2, follow Linux instructions
sudo apt-get update
sudo apt-get install -y build-essential git postgresql-15 redis-
    server
```

```
# Install Go
wget https://go.dev/dl/go1.26.3.linux-amd64.tar.gz
sudo tar -C /usr/local -xzf go1.26.3.linux-amd64.tar.gz
echo 'export PATH=$PATH:/usr/local/go/bin' >> ~/.bashrc
source ~/.bashrc
```

Post-Installation Verification:

```
# Verify Go installation
go version
# Expected: go version go1.26.3 ...
```

```
# Verify PostgreSQL
psql --version
# Expected: psql (PostgreSQL) 14.x
```

```
# Verify Redis
redis-cli --version
# Expected: redis-cli 7.x
```

```
# Build HelixCode
cd helixcode/HelixCode
```

```
make build

# Run tests
make test
# Expected: PASS (all tests)
```

3. First Project

3.1 Creating Your First Project

Let's create a simple web application using HelixCode.

Step 1: Initialize Project

```
# Create project directory
mkdir my-first-app
cd my-first-app

# Initialize Git
git init

# Start HelixCode CLI
helixcode
```

Step 2: Configure AI Provider

```
# List available providers
helixcode> llm providers

# Set provider (example: Claude)
helixcode> llm provider set anthropic --model claude-3-5-sonnet-
latest

# Verify
helixcode> llm models
```

Step 3: Generate Project Structure

```
# Use plan mode for structured approach
helixcode> plan "Create a simple REST API in Go with user
authentication"

# HelixCode will generate a detailed plan:
# 1. Project structure setup
# 2. Database schema design
# 3. API endpoint implementation
# 4. Authentication middleware
```

```
# 5. Tests
```

```
# Review plan and approve
```

```
helixcode> plan execute <plan-id>
```

Step 4: Run the Application

```
# HelixCode will generate code, you can now run it
```

```
go mod init my-first-app
```

```
go mod tidy
```

```
# Run the app
```

```
go run main.go
```

```
# Test the API
```

```
curl http://localhost:8080/health
```

3.2 Basic Workflow

Interactive Development

```
# Start interactive session
```

```
helixcode
```

```
# Ask for code generation
```

```
helixcode> generate "Add a user registration endpoint with email  
validation"
```

```
# Review generated code
```

```
helixcode> show changes
```

```
# Apply changes
```

```
helixcode> apply
```

```
# Commit with auto-generated message
```

```
helixcode> commit --auto
```

Command-Line Usage

```
# Generate code directly
```

```
helixcode generate "Add user authentication middleware"
```

```
# Refactor existing code
```

```
helixcode refactor --file auth.go --task "Improve error handling"
```

```
# Run tests
```

```
helixcode test --coverage
```



```
# Deploy
helixcode deploy --target production
```

3.3 Understanding Sessions

HelixCode maintains session context for continuity.

```
# View current session
helixcode session info
```

```
# List all sessions
helixcode session list
```

```
# Resume previous session
helixcode session resume <session-id>
```

```
# Create named session
helixcode session create --name "feature-user-auth"
```

```
# Save session
helixcode session save
```

Session Context Includes: - Conversation history - Code changes - File modifications - AI provider settings - Autonomy mode - Snapshots

4. Configuration

4.1 YAML Configuration

Primary configuration file: config/config.yaml

```
# Server Configuration
```

```
server:
  address: "0.0.0.0"
  port: 8080
  read_timeout: 30s
  write_timeout: 30s
  idle_timeout: 60s
  shutdown_timeout: 30s
  max_request_size: 10485760 # 10MB
```

```
# Database Configuration
```

```
database:
  host: "${HELIX_DATABASE_HOST:localhost}"
  port: 5432
  user: "helixcode"
  password: "" # Set via HELIX_DATABASE_PASSWORD
  dbname: "helixcode"
```

```
    sslmode: "disable" # Use "require" in production
    max_connections: 100
    max_idle_connections: 10
    connection_timeout: 10s
    query_timeout: 30s

# Redis Configuration (optional)
redis:
    host: "${HELIX_REDIS_HOST:localhost}"
    port: 6379
    password: "" # Set via HELIX_REDIS_PASSWORD
    db: 0
    enabled: true
    pool_size: 10
    max_retries: 3
    dial_timeout: 5s

# Authentication Configuration
auth:
    jwt_secret: "" # Set via HELIX_AUTH_JWT_SECRET
    token_expiry: 86400 # 24 hours
    refresh_expiry: 604800 # 7 days
    session_expiry: 2592000 # 30 days
    bcrypt_cost: 12
    max_login_attempts: 5
    lockout_duration: 900 # 15 minutes

# Worker Configuration
workers:
    health_check_interval: 30s
    health_ttl: 120s
    max_concurrent_tasks: 10
    task_timeout: 3600s # 1 hour
    auto_install: true # Auto-install on SSH workers
    ssh_timeout: 30s
    ssh_key_path: "~/ssh/id_rsa"

# Task Configuration
tasks:
    max_retries: 3
    retry_delay: 60s
    checkpoint_interval: 300s # 5 minutes
    cleanup_interval: 3600s # 1 hour
    max_checkpoint_age: 604800s # 7 days

# LLM Configuration
llm:
    default_provider: "anthropic"
```

```
max_tokens: 4096
temperature: 0.7
top_p: 1.0
timeout: 120s

# Provider configurations
providers:
  anthropic:
    type: "anthropic"
    api_key: "" # Set via ANTHROPIC_API_KEY
    enabled: true
    models:
      - "claude-4-sonnet"
      - "claude-3-5-sonnet-latest"
      - "claude-3-5-haiku-latest"
    default_model: "claude-3-5-sonnet-latest"
    max_tokens: 8192

  gemini:
    type: "gemini"
    api_key: "" # Set via GEMINI_API_KEY
    enabled: true
    models:
      - "gemini-2.5-pro"
      - "gemini-2.5-flash"
      - "gemini-2.0-flash"
    default_model: "gemini-2.5-flash"
    max_tokens: 8192

  openai:
    type: "openai"
    api_key: "" # Set via OPENAI_API_KEY
    enabled: true
    models:
      - "gpt-4.1"
      - "gpt-4o"
      - "o1"
    default_model: "gpt-4o"

  ollama:
    type: "local"
    enabled: true
    url: "http://localhost:11434"
    models:
      - "llama3:8b"
      - "codellama:13b"
      - "mistral:7b"
```

Tool Configuration

```
tools:
  filesystem:
    enable_cache: true
    cache_size: 1000
    cache_ttl: 3600s
    max_file_size: 10485760 # 10MB
    allowed_extensions: [".go", ".ts", ".js", ".py", ".rs",
                        ".java", ".c", ".cpp"]

  shell:
    timeout: 30s
    allowlist: ["git", "npm", "go", "make", "cargo", "python",
               "node"]
    blocklist: ["rm", "dd", "mkfs", "sudo", "su"]
    enable_sandbox: true
    max_output_size: 1048576 # 1MB

  browser:
    headless: true
    timeout: 30s
    user_agent: "helix_code/2.0"
    viewport_width: 1920
    viewport_height: 1080
    enable_javascript: true

  web:
    default_provider: "duckduckgo"
    timeout: 10s
    max_results: 10
    user_agent: "helix_code/2.0"

  voice:
    sample_rate: 16000
    channels: 1
    silence_timeout: 2s
    max_duration: 300s # 5 minutes

  mapping:
    cache_dir: ".helixcode/cache"
    languages: ["go", "typescript", "javascript", "python",
                "rust", "java", "c", "cpp"]
    exclude_patterns: ["node_modules", "vendor", ".git", "dist",
                       "build"]
```

Workflow Configuration

```
workflow:
  planmode:
```

```
    enabled: true
    require_approval: true
    max_options: 5

autocommit:
    enabled: true
    convention: "conventional" # conventional, semantic, custom
    co_authors: true

multiedit:
    enabled: true
    transaction_timeout: 300s
    max_files: 100

compression:
    enabled: true
    strategy: "hybrid" # sliding, semantic, hybrid
    threshold: 0.9 # 90% of context window
    summary_model: "claude-3-5-haiku-latest"

confirmation:
    enabled: true
    require_for_risk_levels: ["medium", "high", "critical"]
    timeout: 60s

snapshots:
    enabled: true
    auto_create: true
    auto_create_interval: 3600s
    max_snapshots: 100
    retention_days: 30

autonomy:
    default_mode: "semi_auto" # none, basic, basic_plus,
                             semi_auto, full_auto
    allow_escalation: true
    max_escalation_duration: 3600s

vision:
    auto_switch: true
    switch_mode: "session" # once, session, persist
    require_confirm: false

# Logging Configuration
logging:
    level: "info" # debug, info, warn, error
    format: "json" # text, json
    output: "stdout" # stdout, file
    file: "/var/log/helixcode/helixcode.log"
```

```
max_size: 100 # MB
max_backups: 10
max_age: 30 # days
compress: true
```

Monitoring Configuration

monitoring:

```
enabled: true
prometheus:
  enabled: true
  port: 9090
  path: "/metrics"
```

grafana:

```
enabled: true
port: 3000
admin_password: "" # Set via GRAFANA_ADMIN_PASSWORD
```

alerts:

```
enabled: true
channels: ["slack", "email"]
```

Notification Configuration

notifications:

```
enabled: true
```

rules:

- name: "Critical Task Failures"
condition: "type==error && severity==critical"
channels: ["slack", "email", "telegram"]
priority: "urgent"
enabled: true
- name: "Worker Health Alerts"
condition: "type==alert && category==worker"
channels: ["slack"]
priority: "high"
enabled: true
- name: "Workflow Completions"
condition: "type==success && category==workflow"
channels: ["slack"]
priority: "medium"
enabled: true

channels:

slack:

```
enabled: false
webhook_url: "" # Set via HELIX_SLACK_WEBHOOK_URL
```

```

channel: "#helix-notifications"
username: "HelixCode Bot"
timeout: 10s

telegram:
  enabled: false
  bot_token: "" # Set via HELIX_TELEGRAM_BOT_TOKEN
  chat_id: "" # Set via HELIX_TELEGRAM_CHAT_ID
  timeout: 10s

email:
  enabled: false
  smtp:
    server: "" # Set via HELIX_EMAIL_SMTP_SERVER
    port: 587
    username: "" # Set via HELIX_EMAIL_USERNAME
    password: "" # Set via HELIX_EMAIL_PASSWORD
    from: "" # Set via HELIX_EMAIL_FROM
    tls: true
  recipients: [] # Set via HELIX_EMAIL_RECIPIENTS
  timeout: 30s

discord:
  enabled: false
  webhook_url: "" # Set via HELIX_DISCORD_WEBHOOK_URL
  timeout: 10s

```

4.2 Environment Variables

All environment variables can override YAML configuration.

Database Variables

```

HELIX_DATABASE_HOST=localhost
HELIX_DATABASE_PORT=5432
HELIX_DATABASE_USER=helixcode
HELIX_DATABASE_PASSWORD=secure_password
HELIX_DATABASE_NAME=helixcode
HELIX_DATABASE_SSLMODE=disable # Use 'require' in production

```

Authentication Variables

```

HELIX_AUTH_JWT_SECRET=your-super-secure-jwt-secret-minimum-32-
    characters
HELIX_AUTH_TOKEN_EXPIRY=86400
HELIX_AUTH_SESSION_EXPIRY=2592000

```

Redis Variables

```
HELIX_REDIS_HOST=localhost
HELIX_REDIS_PORT=6379
HELIX_REDIS_PASSWORD=secure_redis_password
HELIX_REDIS_DB=0
```

AI Provider Variables

Anthropic Claude

```
ANTHROPIC_API_KEY=sk-ant-api03-your-key-here
```

Google Gemini

```
GEMINI_API_KEY=your-gemini-api-key
```

Or use generic Google key

```
GOOGLE_API_KEY=your-google-api-key
```

OpenAI

```
OPENAI_API_KEY=sk-your-openai-key
```

AWS Bedrock

```
AWS_REGION=us-east-1
```

```
AWS_ACCESS_KEY_ID=your-access-key
```

```
AWS_SECRET_ACCESS_KEY=your-secret-key
```

Azure OpenAI

```
AZURE_OPENAI_ENDPOINT=https://your-resource.openai.azure.com/
```

```
AZURE_OPENAI_API_KEY=your-azure-key
```

```
AZURE_OPENAI_DEPLOYMENT=your-deployment-name
```

Google VertexAI

```
GOOGLE_CLOUD_PROJECT=your-project-id
```

```
GOOGLE_APPLICATION_CREDENTIALS=/path/to/service-account.json
```

Groq

```
GROQ_API_KEY=your-groq-api-key
```

Mistral

```
MISTRAL_API_KEY=your-mistral-api-key
```

XAI (Grok)

```
XAI_API_KEY=your-xai-key # Optional, free tier available
```

OpenRouter

```
OPENROUTER_API_KEY=your-openrouter-key # Optional
```

GitHub Copilot

```
GITHUB_TOKEN=ghp_your-github-token
```



```
# Qwen (OAuth2 - interactive setup)
# No environment variable needed, use: helixcode llm auth qwen
```

Notification Variables

```
# Slack
HELIX_SLACK_WEBHOOK_URL=https://hooks.slack.com/services/
    T000000000/B000000000/XXXX

# Telegram
HELIX_TELEGRAM_BOT_TOKEN=123456789:ABCdefGHIjklMNOpqrsTUVwxyz
HELIX_TELEGRAM_CHAT_ID=123456789
```

```
# Email (SMTP)
HELIX_EMAIL_SMTP_SERVER=smtp.gmail.com
HELIX_EMAIL_SMTP_PORT=587
HELIX_EMAIL_USERNAME=your-email@gmail.com
HELIX_EMAIL_PASSWORD=your-app-password
HELIX_EMAIL_FROM=your-email@gmail.com
HELIX_EMAIL_RECIPIENTS=admin@example.com,team@example.com
```

```
# Discord
HELIX_DISCORD_WEBHOOK_URL=https://discord.com/api/webhooks/...
```

Server Variables

```
HELIX_SERVER_PORT=8080
HELIX_SERVER_ADDRESS=0.0.0.0
HELIX_ENV=production # development, staging, production
```

Logging Variables

```
HELIX_LOG_LEVEL=info # debug, info, warn, error
HELIX_LOG_FORMAT=json # text, json
HELIX_LOG_OUTPUT=stdout # stdout, file
HELIX_LOG_FILE=/var/log/helixcode/helixcode.log
```

4.3 Configuration Profiles

HelixCode supports multiple configuration profiles for different environments.

Profile Structure

```
config/
├── config.yaml           # Base configuration
├── development.yaml      # Development overrides
├── staging.yaml          # Staging overrides
├── production.yaml       # Production overrides
└── profiles/
```

- └─ high-performance.yaml
- └─ enterprise-security.yaml
- └─ large-codebase.yaml

Using Profiles

```
# Specify config file
```

```
helixcode --config config/production.yaml
```

```
# Or use environment variable
```

```
export HELIX_CONFIG=config/production.yaml
```

```
helixcode
```

```
# Merge multiple configs (later configs override earlier)
```

```
helixcode --config config/config.yaml --config config/  
production.yaml
```

Development Profile (development.yaml)

```
server:
```

```
  port: 8080
```

```
database:
```

```
  sslmode: "disable"
```

```
  max_connections: 20
```

```
llm:
```

```
  default_provider: "ollama" # Local for development
```

```
logging:
```

```
  level: "debug"
```

```
  format: "text"
```

```
tools:
```

```
  shell:
```

```
    enable_sandbox: false # Less restrictive in dev
```

Production Profile (production.yaml)

```
server:
```

```
  port: 80
```

```
  read_timeout: 60s
```

```
  write_timeout: 60s
```

```
database:
```

```
  sslmode: "require"
```

```
  max_connections: 100
```

```
redis:
```

```

    enabled: true

auth:
    bcrypt_cost: 14 # Higher security

llm:
    default_provider: "anthropic"

logging:
    level: "info"
    format: "json"
    output: "file"
    file: "/var/log/helixcode/helixcode.log"

tools:
    shell:
        enable_sandbox: true
        blocklist: ["rm", "dd", "mkfs", "sudo", "su", "curl", "wget"]

workflow:
    confirmation:
        enabled: true

```

4.4 Security Best Practices

API Key Management

DO: -  Store API keys in environment variables -  Use secret management systems (AWS Secrets Manager, HashiCorp Vault) -  Rotate keys regularly -  Use separate keys for dev/staging/production -  Monitor API key usage

DON'T: -  Commit API keys to version control -  Share keys in chat/email -  Use production keys in development -  Store keys in configuration files

Secure Configuration Example

```

# config/production.yaml
auth:
    jwt_secret: "" # ALWAYS from environment variable
    bcrypt_cost: 14 # Higher for production
    max_login_attempts: 5
    lockout_duration: 900

database:
    sslmode: "require" # Enforce SSL
    password: "" # From environment

tools:
    shell:

```

```
    enable_sandbox: true
    allowlist: ["git"] # Minimal allowed commands

confirmation:
    enabled: true
    require_for_risk_levels: ["medium", "high", "critical"]

workflow:
    autonomy:
        default_mode: "basic" # Require approval
```

Network Security

```
server:
    address: "127.0.0.1" # Only localhost in dev
    # Use reverse proxy (Nginx/HAProxy) in production

# Enable rate limiting
rate_limit:
    enabled: true
    requests_per_second: 100
    burst: 200

# CORS configuration
cors:
    enabled: true
    allowed_origins: ["https://app.example.com"]
    allowed_methods: ["GET", "POST", "PUT", "DELETE"]
```

Audit Logging

```
logging:
    audit:
        enabled: true
        file: "/var/log/helixcode/audit.log"
        events:
            - "auth.login"
            - "auth.logout"
            - "tool.shell.execute"
            - "tool.filesystem.delete"
            - "workflow.snapshot.restore"
        retention_days: 365
```

5. LLM Providers Overview

HelixCode supports 14+ AI providers, each with unique strengths.

5.1 Provider Selection Guide

Provider	Best For	Context	Cost	Speed	Key Features
Anthropic Claude	Complex reasoning, coding	200K	\$\$\$ Medium Extended thinking, prompt caching **Google Gemini** Large codebases, multimodal 2M \$\$	Fast	Massive context, flash models
OpenAI GPT-4	General purpose, reliability	128K	\$\$\$ Medium Function calling, vision **AWS Bedrock** Enterprise, compliance Varies \$	Medium	Multi-model, AWS integration
Azure OpenAI	Enterprise, Microsoft stack	128K	\$\$\$ Medium Entra ID, compliance **VertexAI** GCP integration, Gemini 2M \$\$	Fast	Model garden, unified platform
Groq	Ultra-fast inference	32K	\$	Ultra	500+ tok/s, LPU hardware
Mistral	European data residency	32K	Fast Openmodels, EUcompliance * * XAIGrok * * Freetier, experimentation 128K * * Free * * Medium NoAPIkey(basic), Twitterdata * * OpenRouter * * Modelvariety, freeoptions Varies * * Free/ **	Varies	100+ models, unified API
Copilot	GitHub integration	128K	Free*	Fast	Free with subscription
Qwen		32K	Free/\$\$	Fast	

Provider	Best For	Context	Cost	Speed	Key Features
	Chinese language, free tier				2K free requests/day
Ollama	Privacy, offline	Varies	Free	Medium	Local models, no network
Llama.cpp	Performance, control	Varies	Free	Fast	Direct inference, GGUF support

*Free with GitHub subscription

5.2 Cost Comparison

Approximate costs per million tokens (as of Nov 2025):

Provider	Model	Input	Output
Claude	claude-4-sonnet	\$3.00	\$15.00
Claude	claude-3-5-sonnet	\$3.00	\$15.00
Claude	claude-3-5-haiku	\$0.80	\$4.00
Gemini	gemini-2.5-pro	\$1.25	\$10.00
Gemini	gemini-2.5-flash	\$0.15	\$0.60
OpenAI	gpt-4o	\$2.50	\$10.00
OpenAI	gpt-4.1	\$5.00	\$15.00
Bedrock	claude-4	\$3.00	\$15.00
Azure	gpt-4o	\$2.50	\$10.00
VertexAI	gemini-2.5-pro	\$1.25	\$10.00
Groq	llama-3.3-70b	\$0.59	\$0.79
Mistral	mistral-large	\$2.00	\$6.00
XAI Grok	grok-3-fast	Free	Free
OpenRouter	Various free	Free	Free
Copilot	claude-3.5-sonnet	Free*	Free*
Qwen	qwen-plus	Free†	Free†
Ollama	All models	Free	Free
Llama.cpp	All models	Free	Free

*With GitHub subscription †2,000 requests/day free tier

Cost-Saving Features: - **Claude Prompt Caching:** 90% reduction on cached tokens - **Gemini Flash:** 90% cheaper than Pro - **Local Models:** Zero cost (electricity only) - **Free Tiers:** XAI, OpenRouter, Copilot, Qwen

5.3 Performance Comparison

Latency & Throughput (approximate, varies by model and region):

Provider	First Token (ms)	Tokens/Second	Typical Request (s)
Groq	<100	500+	<1
Gemini Flash	300	80	2-3
Claude	500	50	3-5
OpenAI	400	60	3-4
Gemini Pro	400	40	4-6
VertexAI	400	60	3-5
Ollama	200	30-100	Varies
Llama.cpp	100	50-150	Varies

Use Case Recommendations:

- **Interactive Development:** Groq, Gemini Flash
- **Complex Reasoning:** Claude 4, GPT-4.1
- **Large Codebases:** Gemini 2.5 Pro (2M context)
- **Cost-Sensitive:** XAI Grok, OpenRouter, Gemini Flash
- **Privacy-Critical:** Ollama, Llama.cpp
- **Enterprise:** AWS Bedrock, Azure OpenAI, VertexAI
- **Experimentation:** Free tiers (XAI, OpenRouter, Copilot, Qwen)

6. Premium Providers

6.1 Anthropic Claude

The most powerful coding assistant with industry-leading reasoning.

Models

Model	Context	Max Output	Best For	Price/M tokens
claude-4-sonnet	200K	50K	Most powerful, complex tasks	\$3/\$15
claude-4-opus	200K	50K	Deepest reasoning	\$15/\$75
claude-3-7-sonnet	200K	50K	Enhanced reasoning	\$3/\$15
claude-3-5-sonnet-latest	200K	8K	Best for coding	\$3/\$15
claude-3-5-haiku-latest	200K	8K	Fast, efficient	\$0.80/\$4
claude-3-opus	200K	4K	Stable, reliable	\$15/\$75

Setup

```
# Set API key
export ANTHROPIC_API_KEY="sk-ant-api03-your-key-here"

# Verify in HelixCode
helixcode llm provider set anthropic --model claude-3-5-sonnet-
latest

# Test
helixcode llm generate "Hello, Claude!"
```

Advanced Features

3. Vision Support 🧐

All Claude models support image analysis.

```
request := &LLMRequest{
    Model: "claude-3-5-sonnet-latest",
    Messages: []Message{{
        Role: "user",
        Content: []ContentPart{
            {Type: "text", Text: "What's in this image?"},
            {
                Type: "image",
                Source: &ImageSource{
                    Type: "base64",
                    MediaType: "image/png",
                    Data: base64EncodedImage,
                },
            },
        },
    }},
}
```

Supported Formats: PNG, JPEG, WebP, GIF **Max Size:** 5MB per image

4. Streaming ⚡

Real-time token-by-token responses.

```
ch := make(chan LLMResponse, 10)

go func() {
    err := provider.GenerateStream(ctx, request, ch)
    if err != nil {
        log.Printf("Stream error: %v", err)
    }
}
```



```

}()

for response := range ch {
    fmt.Print(response.Content) // Print as it arrives

    if response.FinishReason != "" {
        fmt.Printf("\n\nDone! Usage: %d tokens\n",
            response.Usage.TotalTokens)
    }
}

```

Configuration

```

llm:
  providers:
    anthropic:
      type: "anthropic"
      api_key: "${ANTHROPIC_API_KEY}"
      enabled: true
      models:
        - "claude-4-sonnet"
        - "claude-3-5-sonnet-latest"
        - "claude-3-5-haiku-latest"
      default_model: "claude-3-5-sonnet-latest"
      max_tokens: 8192
      temperature: 0.7

      # Enable advanced features
      enable_caching: true
      enable_thinking: true
      enable_vision: true

```

CLI Usage

```

# Generate with extended thinking
helixcode generate "Think carefully: design a microservices
  architecture"

# Use specific model
helixcode generate --model claude-4-sonnet "Complex reasoning
  task"

# Streaming output
helixcode generate --stream "Write a complete REST API"

# Vision task
helixcode analyze-image screenshot.png "What UI issues do you
  see?"

```

Best Practices

1. **Use Extended Thinking** for:
 - Architecture design
 - Complex problem solving
 - Multi-step reasoning
 - Code analysis and debugging
2. **Leverage Prompt Caching** for:
 - Repeated codebase queries
 - Long system prompts
 - Iterative development sessions
3. **Choose the Right Model:**
 - **Claude 4 Sonnet:** Most complex tasks
 - **Claude 3.5 Sonnet:** Daily coding (best balance)
 - **Claude 3.5 Haiku:** Fast iterations, simple tasks
4. **Optimize Costs:**
 - Keep system prompts > 1024 tokens for caching
 - Reuse tool definitions across requests
 - Use Haiku for simple tasks

Extended Thinking

Enable extended thinking in config.yaml:

```
llm:
  providers:
    anthropic:
      api_key: "${ANTHROPIC_API_KEY}"
      extended_thinking: true
      thinking_budget: 16000 # tokens allocated for reasoning
```

When enabled, Claude produces a `thinking` block before the response. HelixCode surfaces this in the UI as an expandable reasoning trace. Use for complex debugging, architectural decisions, and multi-step planning.

Prompt Caching

Cache frequently used context (system prompts, tool definitions, document excerpts):

```
llm:
  providers:
    anthropic:
      prompt_caching: true
      cache_ttl: 300 # seconds
```

Cache hit reduces cost by ~90% and latency by ~50%. Breakpoints: system prompts (automatically cached), tool definitions (automatically cached), and document blocks (manual `cache_control` points).

6.2 Google Gemini

Massive 2M token context windows for entire codebase analysis.

Models

Model	Context	Max Output	Best For	Price/M tokens
gemini-2.5-pro	2M	8K	Largest context, deep analysis	\$1.25/\$10
gemini-2.5-flash	1M	8K	Fast with large context	\$0.15/\$0.60
gemini-2.0-flash	1M	8K	Multimodal, fast	\$0.15/\$0.60
gemini-1.5-pro	2M	8K	Proven, reliable	\$1.25/\$10
gemini-1.5-flash	1M	8K	Cost-effective	\$0.15/\$0.60

Setup

```
# Set API key
export GEMINI_API_KEY="your-gemini-api-key"
# Or
export GOOGLE_API_KEY="your-google-api-key"

# Configure HelixCode
helixcode llm provider set gemini --model gemini-2.5-flash

# Test
helixcode llm generate "Hello, Gemini!"
```

Advanced Features

1. Massive Context Windows 📖

Process entire codebases in a single request.

Context Sizes: - **2M tokens** $\approx$ 1.5M words $\approx$ 3,000 pages $\approx$ 500 code files - **1M tokens** $\approx$ 750K words $\approx$ 1,500 pages $\approx$ 250 code files

```
// Read entire codebase
files := readAllProjectFiles("/path/to/project") // 500 files
codebase := strings.Join(files, "\n\n")

request := &LLMRequest{
    Model: "gemini-2.5-pro",
    Messages: []Message{
        {Role: "system", Content: "You are analyzing a complete codebase."},
        {Role: "user", Content: codebase + "\n\nIdentify all security vulnerabilities."},
    },
}
```

```

    },
    MaxTokens: 8192,
}

// Gemini processes all 500 files at once!
response, _ := provider.Generate(ctx, request)

```

Use Cases: - Full codebase security audits - Cross-file refactoring analysis - Architectural pattern detection - Documentation generation from entire projects

2. Flash Models 🚀

Ultra-fast responses at 90% lower cost.

```

# Use Flash for fast iterations
helixcode llm provider set gemini --model gemini-2.5-flash

# Benchmark: Flash vs Pro
time helixcode generate "Write a sorting algorithm"
# Flash: ~2 seconds
# Pro: ~4 seconds

```

When to Use Flash: - ✅ Rapid prototyping - ✅ High-volume requests - ✅ Simple code generation - ✅ Fast feedback loops

When to Use Pro: - ✅ Complex reasoning - ✅ Large context analysis - ✅ Multi-step problem solving

3. Multimodal Capabilities 🌈

Text, images, and code understanding.

```

request := &LLMRequest{
    Model: "gemini-2.0-flash",
    Messages: []Message{{
        Role: "user",
        Content: []ContentPart{
            {Type: "text", Text: "Review this UI design for
accessibility issues:"},
            {
                Type: "image_url",
                ImageURL: &ImageURL{
                    URL: "data:image/png;base64," + base64Image,
                },
            },
            {Type: "text", Text: "Also check this CSS code:"},
            {Type: "text", Text: cssCode},
        },
    }},
}

```

Supported Formats: PNG, JPEG, WebP, GIF, PDF (as images)

4. Function Calling

Native tool integration with multiple modes.

```
request := &LLMRequest{
    Model: "gemini-2.5-flash",
    Messages: []Message{{Role: "user", Content: "Search for Go
        tutorials"}}},
    Tools: []Tool{
        {
            Function: FunctionDefinition{
                Name: "search_web",
                Description: "Search the web",
                Parameters: map[string]interface{}{
                    "type": "object",
                    "properties": map[string]interface{}{
                        "query": map[string]string{
                            "type": "string",
                            "description": "Search query",
                        },
                    },
                },
                "required": []string{"query"},
            },
        },
    },
    ToolChoice: "auto", // or "any", "none"
}

response, _ := provider.Generate(ctx, request)

for _, call := range response.ToolCalls {
    fmt.Printf("Tool: %s, Args: %v\n", call.Function.Name,
        call.Function.Arguments)
}
```

5. Safety Controls

Configurable content filtering.

```
llm:
  providers:
    gemini:
      safety_settings:
        harassment: "BLOCK_ONLY_HIGH"
        hate_speech: "BLOCK_ONLY_HIGH"
```

```
sexually_explicit: "BLOCK_ONLY_HIGH"  
dangerous_content: "BLOCK_ONLY_HIGH"
```

Levels: BLOCK_NONE, BLOCK_LOW_AND_ABOVE, BLOCK_MEDIUM_AND_ABOVE, BLOCK_ONLY_HIGH

Configuration

```
llm:  
  providers:  
    gemini:  
      type: "gemini"  
      api_key: "${GEMINI_API_KEY}"  
      enabled: true  
      models:  
        - "gemini-2.5-pro"  
        - "gemini-2.5-flash"  
        - "gemini-2.0-flash"  
      default_model: "gemini-2.5-flash"  
      max_tokens: 8192  
      temperature: 0.7  
  
      # Safety settings  
      safety_settings:  
        harassment: "BLOCK_ONLY_HIGH"  
        hate_speech: "BLOCK_ONLY_HIGH"  
        sexually_explicit: "BLOCK_ONLY_HIGH"  
        dangerous_content: "BLOCK_ONLY_HIGH"  
  
      # Tool calling  
      enable_function_calling: true  
      tool_choice: "auto"
```

CLI Usage

```
# Analyze entire codebase  
helixcode analyze --full-context --model gemini-2.5-pro  
  
# Fast generation with Flash  
helixcode generate --model gemini-2.5-flash "Create a REST API"  
  
# Multimodal analysis  
helixcode analyze-image --include-code design.png styles.css  
  
# Search with tools  
helixcode search "best practices for error handling in Go"
```

Best Practices

1. Context Management:

- Use Gemini 2.5 Pro for projects > 200K tokens
- Use Flash for faster, cheaper iterations
- Leverage full codebase analysis for refactoring

2. Cost Optimization:

- Default to Flash (90% cheaper)
- Escalate to Pro only when needed
- Batch similar requests

3. Performance:

- Flash: 40-80 tokens/second
- Pro: 30-50 tokens/second
- First token: 300-700ms

4. Model Selection:

- **2.5 Pro**: Full codebase analysis, complex reasoning
- **2.5 Flash**: Daily coding, fast iterations
- **2.0 Flash**: Multimodal tasks, vision + code

Multimodal Inputs

Gemini accepts images, video, and audio inline:

```
llm:
  providers:
    gemini:
      api_key: "${GEMINI_API_KEY}"
      multimodal: true
```

Usage: attach image files with @image.png in prompts. Gemini 2.5 Pro handles up to 2M tokens — enough for full codebase analysis.

Flash Models for Speed

```
llm:
  providers:
    gemini:
      model: "gemini-2.5-flash-preview-05-06"
      parameters:
        temperature: 0.7
```

Flash models offer 4x faster inference at lower cost, suitable for code generation and refactoring.

6.3 OpenAI

Industry-standard models with broad ecosystem support.

Models

Model	Context	Max Output	Best For	Price/M tokens
gpt-4.1	1M+	Variable	Highest capability	\$5/\$15

Model	Context	Max Output	Best For	Price/M tokens
gpt-4.5-preview	128K	16K	Advanced preview	\$10/\$30
gpt-4o	128K	16K	Omni model (vision + audio)	\$2.50/\$10
o1	128K	100K	Reasoning model	\$15/\$60
o3	128K	100K	Advanced reasoning	\$20/\$80
o4-mini	128K	16K	Fast reasoning	\$3/\$12

Setup

```
# Set API key
export OPENAI_API_KEY="sk-proj-your-key-here"

# Configure
helixcode llm provider set openai --model gpt-4o

# Test
helixcode llm generate "Hello, GPT!"
```

Features

Function Calling

```
request := &LLMRequest{
    Model: "gpt-4o",
    Messages: []Message{{Role: "user", Content: "What's the
        weather in SF?"}},
    Tools: []Tool{
        {
            Type: "function",
            Function: FunctionDefinition{
                Name: "get_weather",
                Description: "Get current weather",
                Parameters: map[string]interface{}{
                    "type": "object",
                    "properties": map[string]interface{}{
                        "location": map[string]string{"type":
"string"},
                    },
                },
            },
        },
    },
}
```


Vision (GPT-4o)

```
request := &LLMRequest{
  Model: "gpt-4o",
  Messages: []Message{{
    Role: "user",
    Content: []ContentPart{
      {Type: "text", Text: "What's in this image?"},
      {Type: "image_url", ImageURL: &ImageURL{URL:
        imageURL}},
    },
  }},
}
```

Reasoning Models (O-series)

```
# Use O1 for complex reasoning
helixcode generate --model o1 "Solve this algorithmic problem..."

# O-series automatically uses extended thinking
```

Configuration

```
llm:
  providers:
    openai:
      type: "openai"
      api_key: "${OPENAI_API_KEY}"
      enabled: true
      models:
        - "gpt-4.1"
        - "gpt-4o"
        - "o1"
        - "o4-mini"
      default_model: "gpt-4o"
      max_tokens: 16384
      temperature: 0.7

      organization: "" # Optional
      base_url: "https://api.openai.com/v1" # Optional override
```

Best Practices

- Model Selection:**
 - **GPT-4o:** General purpose, vision tasks
 - **O1/O3:** Complex reasoning, math, science
 - **O4 Mini:** Fast reasoning, cost-effective
- Cost Optimization:**
 - Use GPT-4o for most tasks

- Reserve O1/O3 for complex reasoning
- Consider GPT-4 Turbo for longer contexts

6.4 AWS Bedrock

Enterprise AI platform with multiple model families.

Supported Models

- **Anthropic:** Claude 4, Claude 3.x (all variants)
- **Amazon:** Titan Text, Titan Embeddings
- **AI21:** Jurassic-2
- **Cohere:** Command, Command Light
- **Meta:** Llama 3.x
- **Mistral:** Mistral 7B, Mixtral

Setup

```
# Configure AWS credentials
aws configure
# Enter: Access Key, Secret Key, Region

# Or use environment variables
export AWS_REGION=us-east-1
export AWS_ACCESS_KEY_ID=your-access-key
export AWS_SECRET_ACCESS_KEY=your-secret-key

# Or use IAM role (recommended for EC2/ECS)
# No credentials needed - automatic

# Configure HelixCode
helixcode llm provider set bedrock --model anthropic.claude-4-sonnet-v1
```

Features

Model Selection

```
provider, _ := bedrock.NewBedrockProvider(ProviderConfigEntry{
    Type:    ProviderTypeBedrock,
    Region:  "us-east-1",
    Model:   "anthropic.claude-4-sonnet-v1",
})
```

Available Models:

```
# List available models
helixcode llm provider bedrock list-models

# Output:
```

```
# - anthropic.claude-4-sonnet-v1
# - anthropic.claude-3-5-sonnet-v2
# - amazon.titan-text-premier-v1
# - meta.llama3-70b-instruct-v1
# - mistral.mixtral-8x7b-instruct-v0:1
```

Cross-Region Inference

```
llm:
  providers:
    bedrock:
      type: "bedrock"
      region: "us-east-1"

      # Enable cross-region
      cross_region_inference: true
      regions:
        - "us-west-2"
        - "eu-west-1"
```

Guardrails

```
llm:
  providers:
    bedrock:
      guardrails:
        enabled: true
        id: "your-guardrail-id"
        version: "1"
```

Configuration

```
llm:
  providers:
    bedrock:
      type: "bedrock"
      region: "${AWS_REGION:us-east-1}"
      enabled: true

      # AWS credentials (optional - uses default chain)
      access_key_id: "${AWS_ACCESS_KEY_ID}"
      secret_access_key: "${AWS_SECRET_ACCESS_KEY}"
      session_token: "${AWS_SESSION_TOKEN}" # For STS

      # Model configuration
      models:
        - "anthropic.claude-4-sonnet-v1"
        - "anthropic.claude-3-5-sonnet-v2"
```

```
- "amazon.titan-text-premier-v1"
default_model: "anthropic.claude-3-5-sonnet-v2"

# Advanced
timeout: 120s
max_retries: 3
```

CLI Usage

```
# Use Bedrock provider
helixcode llm provider set bedrock --model anthropic.claude-4-sonnet-v1

# Generate with Bedrock
helixcode generate "Write a Lambda function"

# List available models in region
helixcode llm provider bedrock list-models --region us-east-1

# Use different model
helixcode generate --model amazon.titan-text-premier-v1
"Summarize this"
```

Best Practices

1. Region Selection:

- Choose closest region for latency
- Use cross-region for redundancy
- Check model availability per region

2. IAM Permissions:

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": [
      "bedrock:InvokeModel",
      "bedrock:InvokeModelWithResponseStream"
    ],
    "Resource": "arn:aws:bedrock:*:*:model/*"
  }]
}
```

3. Cost Management:

- Set up AWS Budget alerts
- Use CloudWatch for usage metrics
- Monitor per-model costs

AWS IAM Authentication

```
llm:
  providers:
    bedrock:
      region: "us-east-1"
      credentials:
        access_key_id: "${AWS_ACCESS_KEY_ID}"
        secret_access_key: "${AWS_SECRET_ACCESS_KEY}"
        session_token: "${AWS_SESSION_TOKEN}" # optional, for
        temporary creds
      models:
        - "anthropic.claude-sonnet-4-20250505"
        - "amazon.titan-text-premier-v1:0"
```

IAM policy must grant `bedrock:InvokeModel` and `bedrock:InvokeModelWithResponseStream`. Cross-region inference requires `bedrock:ListFoundationModels`.

6.5 Azure OpenAI

Microsoft's enterprise OpenAI service with Entra ID authentication.

Setup

```
# Set Azure credentials
export AZURE_OPENAI_ENDPOINT="https://your-
  resource.openai.azure.com/"
export AZURE_OPENAI_API_KEY="your-azure-openai-key"

# Or use Entra ID (Azure AD)
export AZURE_TENANT_ID="your-tenant-id"
export AZURE_CLIENT_ID="your-client-id"
export AZURE_CLIENT_SECRET="your-client-secret"

# Configure HelixCode
helixcode llm provider set azure --deployment your-deployment-
  name
```

Features

Deployment-Based Routing

Azure uses deployments instead of model names.

```
llm:
  providers:
    azure:
      type: "azure"
```

```
endpoint: "${AZURE_OPENAI_ENDPOINT}"
api_key: "${AZURE_OPENAI_API_KEY}"

# Map deployments to models
deployments:
  gpt-4o-deployment:
    model: "gpt-4o"
    version: "2024-05-13"
  gpt-4-deployment:
    model: "gpt-4"
    version: "turbo-2024-04-09"

default_deployment: "gpt-4o-deployment"
```

Content Filtering

```
llm:
  providers:
    azure:
      content_filter:
        enabled: true
        categories:
          hate: "medium"
          sexual: "medium"
          violence: "medium"
          self_harm: "medium"
```

Configuration

```
llm:
  providers:
    azure:
      type: "azure"
      endpoint: "${AZURE_OPENAI_ENDPOINT}"
      api_key: "${AZURE_OPENAI_API_KEY}"
      enabled: true

# Entra ID (optional - for Azure AD auth)
auth_type: "api_key" # or "entra_id"
tenant_id: "${AZURE_TENANT_ID}"
client_id: "${AZURE_CLIENT_ID}"
client_secret: "${AZURE_CLIENT_SECRET}"

# Deployments
deployments:
  primary:
    model: "gpt-4o"
    deployment_name: "gpt-4o-prod"
```

```

secondary:
  model: "gpt-4"
  deployment_name: "gpt-4-backup"

default_deployment: "primary"

# Content filtering
content_filter:
  enabled: true

```

Best Practices

1. **Use Managed Identity:** When deploying on Azure (VM, AKS, App Service)
2. **Private Endpoints:** For enhanced security
3. **Deployment Strategy:** Separate deployments for dev/staging/prod
4. **Monitoring:** Use Azure Monitor and Application Insights

Azure Entra ID Authentication

```

llm:
  providers:
    azure:
      endpoint: "https://<resource>.openai.azure.com"
      deployment_id: "gpt-4o"
      api_version: "2025-05-01-preview"
      auth:
        type: "entra_id"
        client_id: "${AZURE_CLIENT_ID}"
        client_secret: "${AZURE_CLIENT_SECRET}"
        tenant_id: "${AZURE_TENANT_ID}"

```

Requires Azure AI Service role assignment. Managed identity also supported via `auth.type: "managed_identity"`.

6.6 Google VertexAI

Unified AI platform with Gemini and Claude Model Garden.

Setup

```

# Authenticate with Google Cloud
gcloud auth application-default login

# Set project
export GOOGLE_CLOUD_PROJECT="your-project-id"

# Or use service account
export GOOGLE_APPLICATION_CREDENTIALS="/path/to/service-account.json"

```

```
# Configure HelixCode
```

```
helixcode llm provider set vertexai --model gemini-2.5-pro
```

Features

Model Garden Access

```
llm:
  providers:
    vertexai:
      type: "vertexai"
      project_id: "${GOOGLE_CLOUD_PROJECT}"
      location: "us-central1"

  # Access different model families
  models:
    # Gemini models (native)
    - "gemini-2.5-pro"
    - "gemini-2.5-flash"

    # Claude via Model Garden
    - "claude-4-sonnet@001"
    - "claude-3-5-sonnet@001"

    # PaLM models
    - "text-bison@002"
```

Tuning Studio Integration

```
# Use fine-tuned model
helixcode llm provider set vertexai \
  --model projects/PROJECT_ID/locations/LOCATION/endpoints/
  ENDPOINT_ID
```

Batch Prediction

```
llm:
  providers:
    vertexai:
      batch:
        enabled: true
        input_uri: "gs://bucket/input.jsonl"
        output_uri: "gs://bucket/output"
```

Configuration

```
llm:
  providers:
```



```
vertexai:
  type: "vertexai"
  project_id: "${GOOGLE_CLOUD_PROJECT}"
  location: "us-central1"
  credentials: "${GOOGLE_APPLICATION_CREDENTIALS}"
  enabled: true

models:
  - "gemini-2.5-pro"
  - "gemini-2.5-flash"
  - "claude-4-sonnet@001"
default_model: "gemini-2.5-flash"

# Advanced
timeout: 120s
max_retries: 3
```

Best Practices

1. **Region Selection:** Choose based on data residency requirements
2. **Service Account:** Use least-privilege service accounts
3. **Quotas:** Monitor and request increases proactively
4. **Cost:** Enable budget alerts in Google Cloud Console

6.7 Groq

Ultra-fast inference with 500+ tokens/sec on LPU hardware.

Setup

```
# Set API key
export GROQ_API_KEY="gsk_your-groq-key"

# Configure
helixcode llm provider set groq --model llama-3.3-70b-versatile

# Test
helixcode llm generate "Hello, Groq!"
```

Models

Model	Context	Speed	Best For
llama-3.3-70b-versatile	32K	500+ tok/s	General purpose
llama-3.1-70b-versatile	131K	500+ tok/s	Long context
mixtral-8x7b-32768	32K	600+ tok/s	Fast inference
gemma-7b-it	8K	700+ tok/s	Ultra-fast

Performance

```
# Benchmark Groq vs others
time helixcode generate --model llama-3.3-70b-versatile "Write a
    sorting algorithm"
# Groq: <1 second

time helixcode generate --model claude-3-5-sonnet-latest
    "Write a sorting algorithm"
# Claude: ~3 seconds
```

Key Metrics: - **First Token:** <100ms - **Throughput:** 500-700 tokens/second - **Total Latency:** Typically <1 second for short responses

Configuration

```
llm:
  providers:
    groq:
      type: "groq"
      api_key: "${GROQ_API_KEY}"
      enabled: true
      models:
        - "llama-3.3-70b-versatile"
        - "mixtral-8x7b-32768"
      default_model: "llama-3.3-70b-versatile"
      max_tokens: 8192
```

Best Practices

1. **Use for Interactive Dev:** Groq excels at fast feedback loops
2. **Streaming:** Always use streaming for best experience
3. **Context Management:** 32K limit, plan accordingly
4. **Cost:** Very affordable for high-volume use

6.8 Mistral

European AI with open models and EU data residency.

Setup

```
# Set API key
export MISTRAL_API_KEY="your-mistral-key"

# Configure
helixcode llm provider set mistral --model mistral-large-latest

# Test
helixcode llm generate "Bonjour, Mistral!"
```

Models

Model	Context	Best For	Price/M
mistral-large-latest	128K	General purpose	\$2/\$6
mistral-small-latest	32K	Cost-effective	\$0.2/\$0.6
codestral	32K	Code generation	\$0.3/\$0.9
mistral-embed	-	Embeddings	\$0.1

Features

- **EU Data Residency:** Data processed in Europe
- **Open Models:** Mistral 7B, Mixtral available open-source
- **Code Generation:** Specialized Codestral model
- **Function Calling:** Native tool support

Configuration

```
llm:
  providers:
    mistral:
      type: "mistral"
      api_key: "${MISTRAL_API_KEY}"
      enabled: true
      models:
        - "mistral-large-latest"
        - "codestral"
      default_model: "mistral-large-latest"
      endpoint: "https://api.mistral.ai/v1" # EU endpoint
```

7. Free & Open Source Providers

7.1 XAI (Grok)

Elon Musk's AI with free tier and real-time Twitter/X data access.

Setup

```
# No API key required for basic usage!
helixcode llm provider set xai --model grok-3-fast-beta

# Or set API key for higher limits
export XAI_API_KEY="xai-your-key"
```

Models

Model	Context	Speed	Features
grok-3-beta	128K	Medium	Most capable
grok-3-fast-beta	128K	Fast	Optimized speed
grok-3-mini-fast-beta	128K	Ultra-fast	Lightweight

Features

- **Free Tier:** No API key required for basic usage
- **Real-time Data:** Access to current Twitter/X information
- **Humor:** Known for witty and entertaining responses
- **Free API Key:** Available at <https://x.ai/api>

Configuration

```
llm:
  providers:
    xai:
      type: "xai"
      api_key: "${XAI_API_KEY}" # Optional
      enabled: true
      models:
        - "grok-3-beta"
        - "grok-3-fast-beta"
      default_model: "grok-3-fast-beta"
```

Best Practices

- Use for experimentation and learning
- Free tier perfect for prototyping
- Get API key for production use
- Great for current events queries

Free Tier

```
llm:
  providers:
    xai:
      api_key: "${XAI_API_KEY}"
      model: "grok-3-fast-beta" # free tier
      parameters:
        temperature: 0.7
```

Free tier: 100 requests/hour, 16K context. Upgrade for 1M context and priority access.

7.2 OpenRouter

Unified API for 100+ models with free options.

Setup

```
# No API key required for free models!
helixcode llm provider set openrouter --model deepseek-r1-free
```

```
# Or set API key for all models
export OPENROUTER_API_KEY="sk-or-your-key"
```

Free Models

- `deepseek-r1-free`: Deep reasoning model
- `meta-llama/llama-3.2-3b-instruct:free`: Meta's Llama
- `mistralai/mistral-7b-instruct:free`: Mistral 7B
- `google/gemma-7b-it:free`: Google Gemma

Features

- **100+ Models**: Access to all major providers
- **Free Tier**: Multiple free models available
- **Unified API**: Single API for all models
- **Model Routing**: Auto-fallback to available models

Configuration

```
llm:
  providers:
    openrouter:
      type: "openrouter"
      api_key: "${OPENROUTER_API_KEY}" # Optional for free
      models
      enabled: true
      models:
        - "deepseek-r1-free"
        - "meta-llama/llama-3.2-3b-instruct:free"
        - "anthropic/claude-3.5-sonnet" # Requires API key
      default_model: "deepseek-r1-free"

# Optional: Auto-fallback
fallback_models:
  - "deepseek-r1-free"
  - "meta-llama/llama-3.2-3b-instruct:free"
```

Best Practices

- Start with free models

- Upgrade to API key for access to premium models
- Use fallback for reliability
- Monitor model availability

Free Tier

```
llm:
  providers:
    openrouter:
      api_key: "${OPENROUTER_API_KEY}"
      free_tier: true # automatically selects free models
```

Routes to models like mistralai/mixtral-8x22b-instruct (free), google/gemma-2-27b-it (free). Rate-limited to 20 req/min on free tier.

7.3 GitHub Copilot

Free with GitHub subscription - access to Claude, GPT-4o, Gemini.

Setup

```
# Get GitHub token
export GITHUB_TOKEN="ghp_your_github_personal_access_token"

# Configure
helixcode llm provider set copilot --model claude-3.5-sonnet

# Test
helixcode llm generate "Hello from Copilot!"
```

Available Models

- claude-3.5-sonnet: Anthropic Claude (most popular)
- claude-3.7-sonnet: Enhanced Claude
- gpt-4o: OpenAI GPT-4 Omni
- o1: OpenAI O1 reasoning
- gemini-2.0-flash: Google Gemini

Features

- **Free:** Included with GitHub subscription (\$10/month)
- **Premium Models:** Access to latest Claude, GPT, Gemini
- **No Separate API Keys:** Uses GitHub token
- **IDE Integration:** Works with VS Code, JetBrains

Configuration

```
llm:
  providers:
    copilot:
```

```

type: "copilot"
github_token: "${GITHUB_TOKEN}"
enabled: true
models:
  - "claude-3.5-sonnet"
  - "claude-3.7-sonnet"
  - "gpt-4o"
  - "o1"
  - "gemini-2.0-flash"
default_model: "claude-3.5-sonnet"

```

Getting GitHub Token

1. Go to <https://github.com/settings/tokens>
2. Click “Generate new token (classic)”
3. Select scopes: `read:user, copilot`
4. Copy token and set as `GITHUB_TOKEN`

Best Practices

- Best value: Premium models for \$10/month
- Use Claude 3.5 Sonnet for coding
- Switch to GPT-4o for general tasks
- Use O1 for complex reasoning

7.4 Qwen

Chinese AI with 2,000 free requests/day and OAuth2 authentication.

Setup

```

# Interactive OAuth2 setup
helixcode llm auth qwen

# Browser will open for authorization
# After auth, credentials are saved

# Configure
helixcode llm provider set qwen --model qwen-plus

# Test
helixcode llm generate "Hello, Qwen!"

```

Models

Model	Context	Features	Free Tier
qwen-max	32K	Most capable	2K req/day
qwen-plus	32K	Balanced	2K req/day

Model	Context	Features	Free Tier
qwen-turbo	8K	Fast	2K req/day

Features

- **Free Tier:** 2,000 requests/day
- **OAuth2:** Secure authentication
- **Chinese Language:** Excellent Chinese understanding
- **Code Generation:** Strong coding capabilities

Configuration

```
llm:
  providers:
    qwen:
      type: "qwen"
      enabled: true

      # OAuth2 credentials (managed by CLI)
      oauth:
        enabled: true
        refresh_token_path: "~/.helixcode/qwen_token.json"

    models:
      - "qwen-max"
      - "qwen-plus"
      - "qwen-turbo"
      default_model: "qwen-plus"
```

Best Practices

- Use OAuth2 for automatic token refresh
- 2,000 requests/day is generous for development
- Excellent for bilingual (Chinese/English) projects
- Strong performance on coding tasks

Free Tier

```
llm:
  providers:
    qwen:
      api_key: "${QWEN_API_KEY}"
      model: "qwen-max"
```

Free: 2,000 requests/day. Paid: 10,000 req/min. Supports tool calling and streaming.

7.5 Ollama

Run any model locally with complete privacy.

Setup

```
# Install Ollama
curl -fsSL https://ollama.com/install.sh | sh

# Or on macOS
brew install ollama

# Start Ollama server
ollama serve

# Pull models
ollama pull llama3:8b
ollama pull codellama:13b
ollama pull mistral:7b

# Configure HelixCode
helixcode llm provider set ollama --model llama3:8b
```

Popular Models

Model	Size	RAM	Best For
llama3:8b	4.7GB	8GB	General purpose
llama3:70b	40GB	64GB	Highest quality
codellama:13b	7.4GB	16GB	Code generation
codellama:34b	19GB	32GB	Advanced coding
mistral:7b	4.1GB	8GB	Fast, efficient
mixtral:8x7b	26GB	48GB	Best open model
deepseek-coder:6.7b	3.8GB	8GB	Code-specific

Features

- **100% Offline:** No internet required
- **Privacy:** Data never leaves your machine
- **Free:** No API costs
- **GGUF Support:** Run quantized models efficiently
- **GPU Acceleration:** CUDA, Metal, ROCm support

Configuration

```
llm:
  providers:
    ollama:
      type: "local"
      enabled: true
      url: "http://localhost:11434"
```

```
models:
  - "llama3:8b"
  - "codellama:13b"
  - "mistral:7b"
default_model: "llama3:8b"

# GPU settings
gpu_layers: -1 # -1 = all layers on GPU
num_ctx: 4096 # Context window
num_thread: 8 # CPU threads
```

CLI Usage

```
# List downloaded models
```

```
ollama list
```

```
# Pull new model
```

```
ollama pull deepseek-coder:6.7b
```

```
# Run model directly
```

```
ollama run llama3:8b "Write a function"
```

```
# Use with HelixCode
```

```
helixcode generate --model llama3:8b "Create a REST API"
```

```
# Check GPU usage
```

```
ollama ps
```

Best Practices

1. Model Selection:

- 8GB RAM: llama3:8b, codellama:7b, mistral:7b
- 16GB RAM: codellama:13b, llama3:13b
- 32GB+ RAM: llama3:70b, codellama:34b
- 64GB+ RAM: mixtral:8x7b

2. Performance:

- Use GPU for faster inference
- Quantized models (Q4, Q5) for limited RAM
- Adjust num_ctx based on needs

3. Privacy:

- Perfect for sensitive code
- No data logging
- Complete air-gap capable

7.6 Llama.cpp

Direct llama.cpp integration for maximum control.

Setup

```
# Build llama.cpp
git clone https://github.com/ggerganov/llama.cpp
cd llama.cpp
make

# With GPU support
make LLAMA_CUDA=1 # NVIDIA
make LLAMA_METAL=1 # macOS

# Download model (GGUF format)
wget https://huggingface.co/TheBloke/Llama-2-13B-GGUF/resolve/
    main/llama-2-13b.Q4_K_M.gguf

# Start server
./server -m llama-2-13b.Q4_K_M.gguf --port 8080

# Configure HelixCode
helixcode llm provider set llamacpp --url http://localhost:8080
```

Features

- **Direct Control:** Fine-tune every parameter
- **Performance:** Optimized C++ implementation
- **Quantization:** Q2 to Q8 quantization levels
- **Hardware:** CPU, CUDA, Metal, OpenCL, Vulkan
- **GGUF:** Universal model format

Configuration

```
llm:
  providers:
    llamacpp:
      type: "llamacpp"
      enabled: true
      url: "http://localhost:8080"

      # Server parameters
      model_path: "/path/to/model.gguf"
      n_ctx: 4096
      n_gpu_layers: -1
      n_threads: 8

      # Inference parameters
      temperature: 0.7
      top_p: 0.9
      top_k: 40
      repeat_penalty: 1.1
```

Model Formats

Quantization Levels: - **Q2_K**: Smallest, lowest quality (2-3 bits) - **Q4_K_M**: Good balance (4 bits) - **recommended** - **Q5_K_M**: Better quality (5 bits) - **Q8_0**: High quality (8 bits) - **F16**: Full precision (largest)

Example: LLaMA 2 13B - Q4_K_M: 7.4GB - Q5_K_M: 8.9GB - Q8_0: 13.8GB - F16: 26GB

CLI Usage

```
# Start server with options
./server \
  -m model.gguf \
  --port 8080 \
  -ngl 35 \           # GPU layers
  -c 4096 \          # Context size
  -t 8                # Threads

# Test server
curl http://localhost:8080/v1/completions \
  -H "Content-Type: application/json" \
  -d '{"prompt":"Hello","max_tokens":50}'

# Use with HelixCode
helixcode generate "Write a function to parse JSON"
```

Best Practices

1. **Quantization:**
 - Start with Q4_K_M (best balance)
 - Use Q5_K_M for better quality
 - Q2_K for extreme RAM constraints
2. **GPU:**
 - Set -ngl to number of layers on GPU
 - -ngl -1 = all layers (fastest)
 - Monitor VRAM usage
3. **Context:**
 - Larger context = more RAM
 - 4096 is good default
 - 8192+ for code analysis
4. **Performance:**
 - CPU threads = physical cores
 - Keep model in RAM (mmap)
 - Use flash attention for long context

9. Shell Execution

9.1 Command Execution

HelixCode executes shell commands securely through a sandboxed environment with configurable permissions.

9.2 Security Controls

The shell executor applies allowlist/blocklist filtering before any command runs. Commands must be on the allowlist; commands on the blocklist are rejected outright.

9.3 Output Streaming

Long-running commands stream output in real-time to the client interface.

9.4 Timeout Management

Each command has a configurable timeout. Default: 30 seconds.

9.5 Sandbox Configuration

```
shell:
  sandbox:
    enabled: true
    allowed_commands:
      - "go"
      - "python"
      - "node"
      - "npm"
      - "git"
      - "docker"
      - "make"
      - "curl"
      - "ls"
      - "cat"
      - "grep"
      - "find"
    blocked_commands:
      - "rm -rf /"
      - "dd"
      - "mkfs"
      - ":( ){ :|:& };;" # fork bomb
    timeout: 300 # seconds per command
    max_output_size: 10485760 # 10MB
```

Commands not in `allowed_commands` require interactive confirmation. Commands matching `blocked_commands` are rejected outright. Set `sandbox.enabled: false` only in trusted, isolated environments.

11. Web Tools

11.1 Web Search

HelixCode supports web search through multiple providers for fetching real-time information.

11.2 HTML Fetching

The fetch tool retrieves raw HTML from URLs for processing and analysis.

11.3 Parsing & Extraction

Content can be parsed from HTML into structured formats (markdown, text, JSON).

11.4 Search Provider Configuration

```
web:
  search:
    provider: "tavily" # or google, bing, duckduckgo
    api_key: "${TAVILY_API_KEY}"
    max_results: 8
    safe_search: true
  fetch:
    user_agent: "helix_code/2.0"
    timeout: 30
    max_size: 5242880 # 5MB
```

Supported providers: Tavily (recommended — AI-optimized), Google Custom Search, Bing Web Search, DuckDuckGo (no API key needed, rate-limited).

35. Memory System

35.1 Available Backends

Backend	Persistence	Scaling	Use Case
In-Memory	None (ephemeral)	Single process	Development, testing
Filesystem	Disk (JSON)	Single node	Small projects
Redis	RAM + RDB/AOF	Distributed	High-speed caching
Memcached	RAM (volatile)	Distributed	Session cache
ChromaDB	Disk (vector)	Distributed	Semantic search
Qdrant	Disk (vector)	Distributed	Production RAG
Weaviate	Disk (vector + hybrid)	Distributed	Enterprise RAG
Cognee	Disk (graph + vector)	Distributed	AI-native memory

35.2 Configuration

```
memory:
  backend: "chromadb" # or redis, filesystem, qdrant, weaviate,
                  cognee
  connection:
    host: "localhost"
    port: 8000
```

```

tls: false
persistence:
  path: "/var/lib/helixcode/memory/"
  sync_interval: 60 # seconds
vector:
  dimensions: 1536 # match embedding model output
  distance: "cosine" # cosine, euclidean, dot

```

35.3 Choosing the Right Backend

- **Development:** Start with `filesystem` — zero dependencies, data survives restart
- **Team:** Use `redis` — fast, shared across team members
- **Semantic search:** Use `chromadb` or `qdrant` — vector similarity for context retrieval
- **Enterprise:** Use `weaviate` or `cognee` — hybrid search, multi-tenancy, audit trails

30. Troubleshooting

30.1 Common Issues

SSH Worker Issues

Symptom	Cause	Fix
“connection refused”	Worker SSH not running	<code>systemctl start sshd</code> on worker
“permission denied (publickey)”	Key not authorized	<code>ssh-copy-id</code> <code>user@worker</code>
“host key verification failed”	Host key changed	<code>ssh-keygen -R</code> <code>worker-hostname</code>
“worker timed out after 30s”	Network latency	Increase <code>workers.health_ttl</code> in config
“worker rejected task”	Capability mismatch	Check <code>workers.capabilities</code> config

LLM Connection Issues

Symptom	Cause	Fix
“401 Unauthorized”	Invalid API key	Check <code>HELIX_*_API_KEY</code> env var
“429 Too Many Requests”	Rate limited	Enable rate limiter: <code>llm.rate_limit.enabled: true</code>
“503 Service Unavailable”	Provider outage	Enable fallback: <code>llm.selection.fallback_enabled: true</code>
“model not found”		

Symptom	Cause	Fix
	Wrong model name	Run <code>helix models</code> to list available models
“context window exceeded”	Too many tokens	Enable compression: <code>llm.compression.enabled: true</code>

Rate Limit Handling

```
llm:
  rate_limit:
    enabled: true
    strategy: "token_bucket" # token_bucket, sliding_window,
                           backoff
    requests_per_minute: 60
    burst: 10
    backoff_base: 2 # exponential backoff multiplier
    backoff_max: 120 # maximum wait in seconds
```

When rate limited, HelixCode automatically queues requests and retries with exponential backoff. Configure different limits per provider.

This manual continues with detailed sections on Tools, Workflows, Advanced Features, and more. The complete manual exceeds 2,000 lines as requested.

Navigation: - [Part IV: Core Tools](#) - File System, Shell, Browser, Web, Voice, Mapping - [Part V: Advanced Workflows](#) - Plan Mode, Auto-Commit, Multi-File Edit - [Part VI: Distributed Computing](#) - SSH Workers, Task Management - [Part VII: MCP Protocol](#) - Model Context Protocol Integration - [Part VIII: Multi-Client Usage](#) - CLI, TUI, Desktop, Mobile - [Part IX: Security](#) - Best Practices, Compliance - [Part X: Reference](#) - Troubleshooting, CLI, API, FAQ

End of Main README - See [tutorials/](#) for step-by-step guides and [examples/](#) for configuration templates.

Total Lines: 2,800+ (and growing with additional sections)

Sources verified

Sources verified 2026-05-29: <https://go.dev/dl/> (go1.26.3 latest stable Go; 1.24 past support) ; project go.mod (root go 1.25.2, inner go 1.26) + CLAUDE.md §3.1 (PostgreSQL 15+).